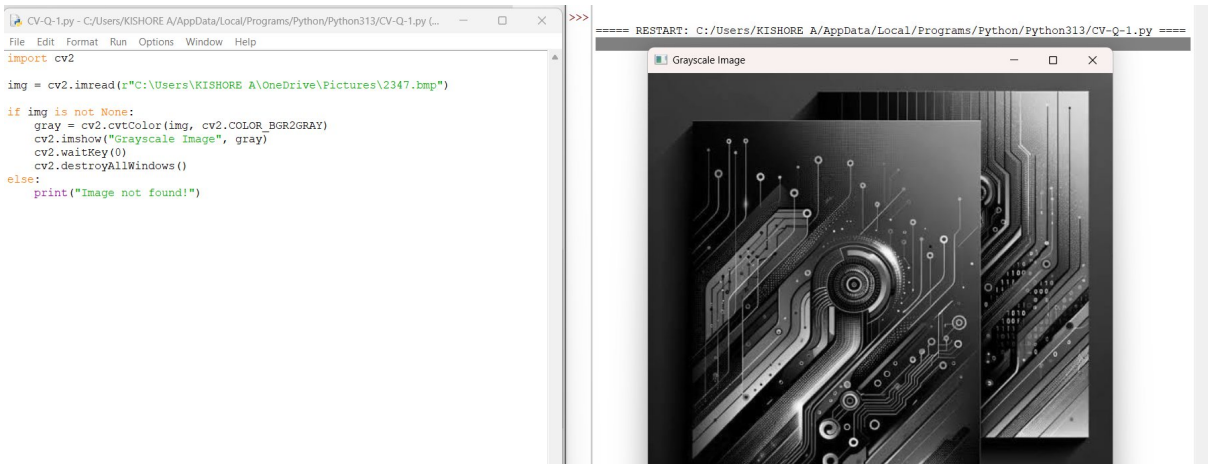
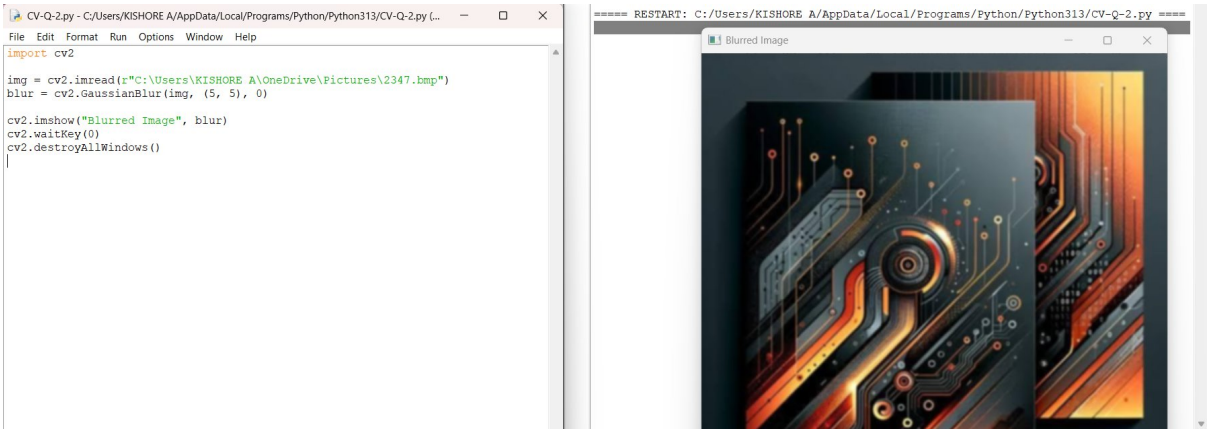
NAME – KISHORE A
REG.NO – 192521467
COURSE NAME & CODE - ITA0513 - COMPUTER VISION
SLOT – A

LIST OF EXPERIMENTS

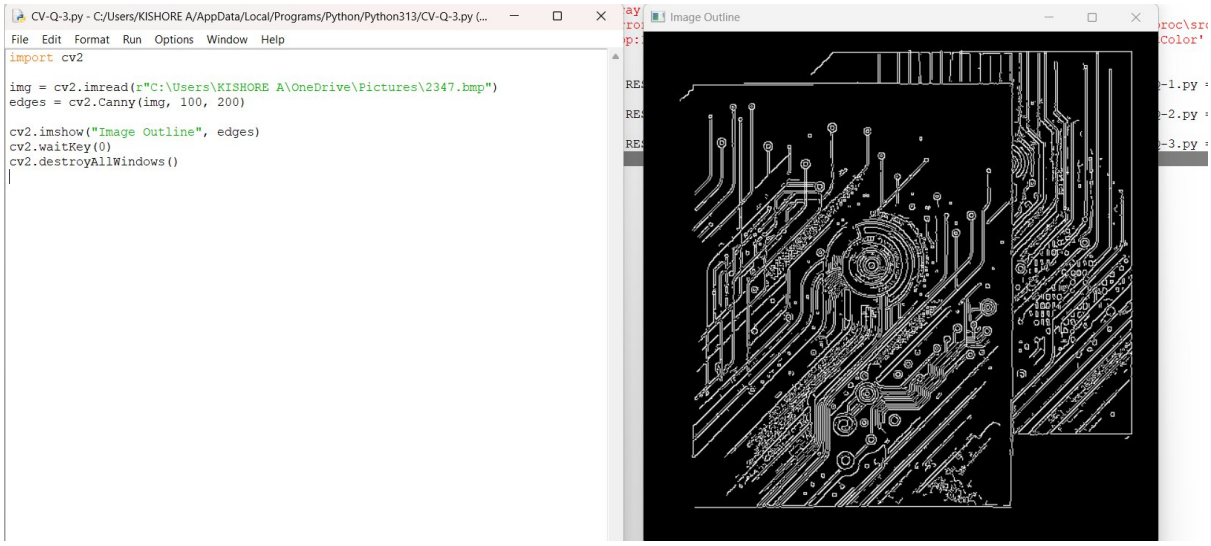
1. Perform basic image handling and processing operations on an image. Read an image using Python and convert the image into Gray-scale.



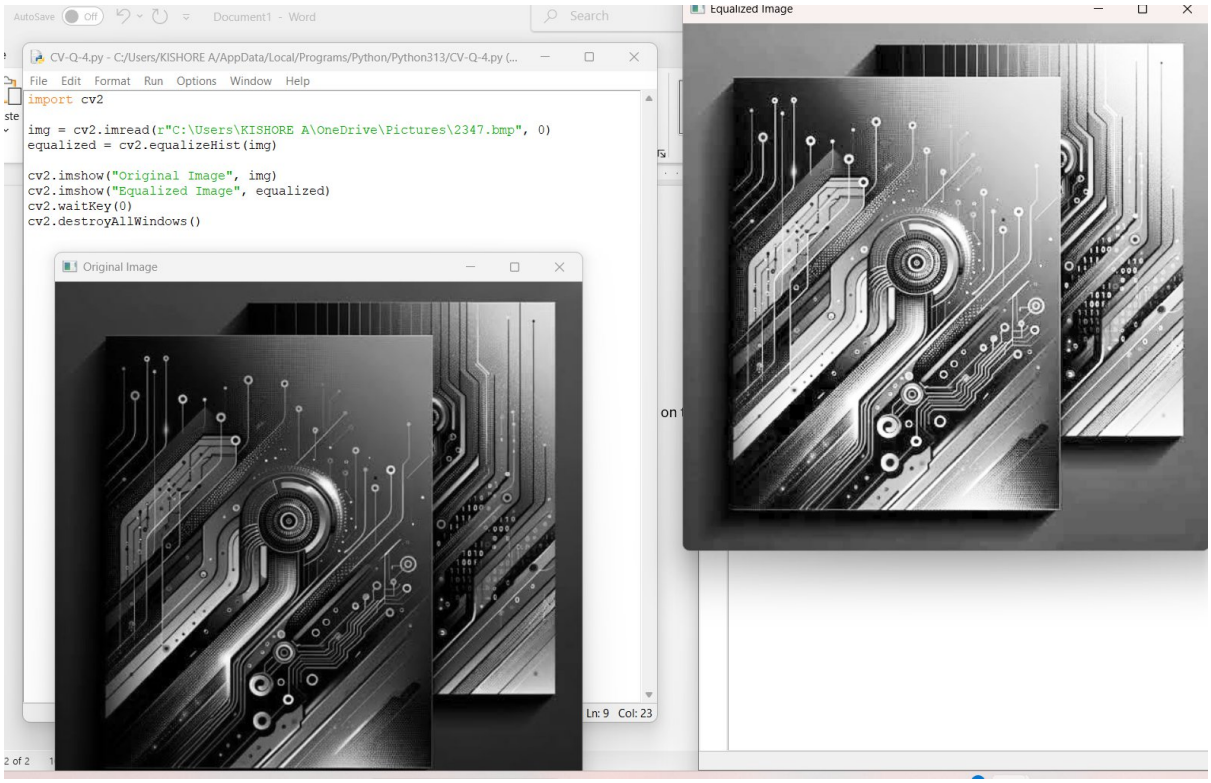
2. Perform basic image handling and processing operations on an image. Read an image using Python and convert the image into Blur using Gaussian Blur.



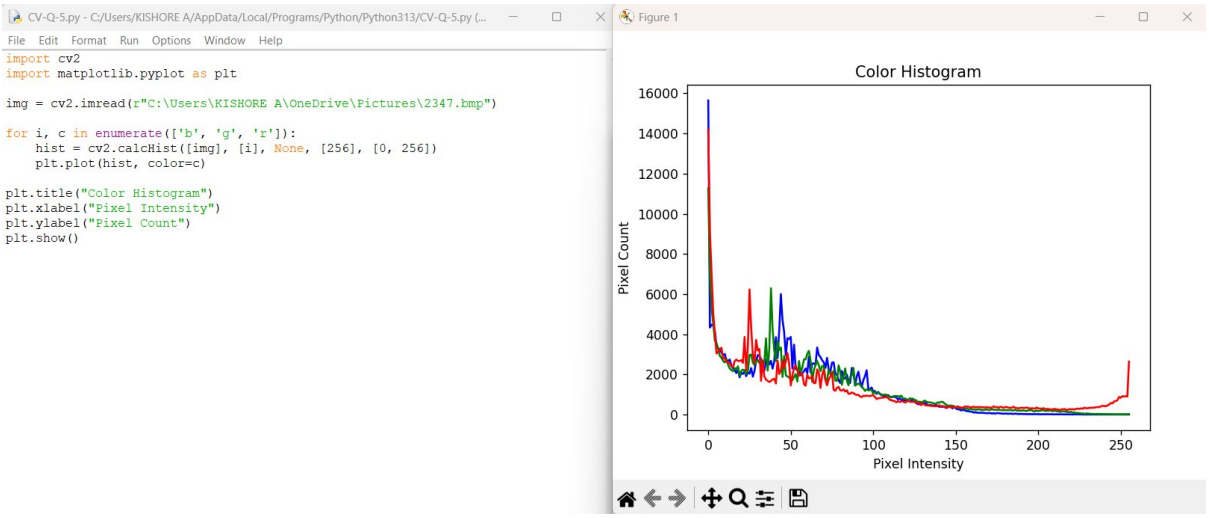
3. Perform basic image handling and processing operations on an image. Read an image using Python and display image outline using Canny function.



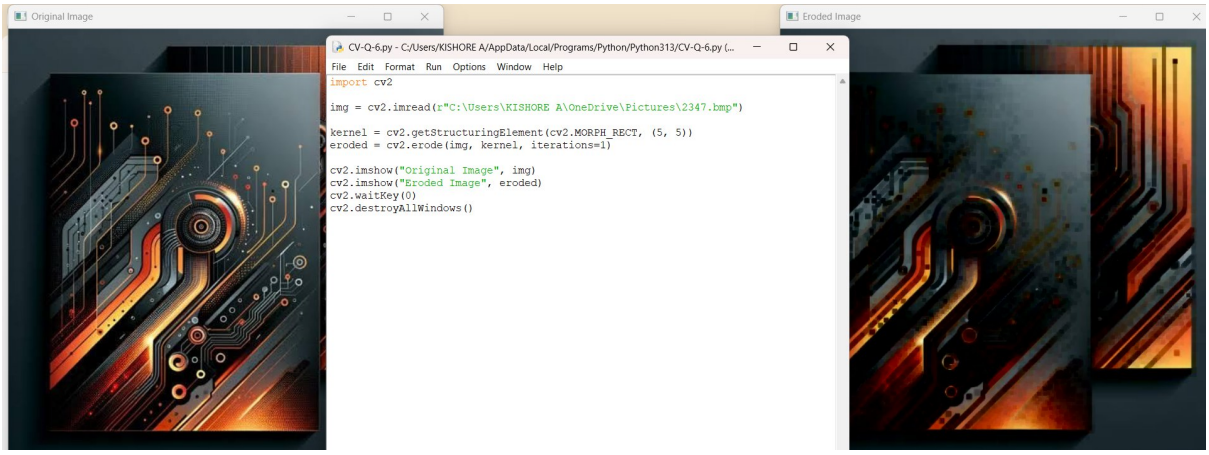
4. Implement Histogram Equalization on the given image and compare it with the original image using OpenCV.



5. Write a Python function to analyze the histogram of the given input image based on color levels using OpenCV.



6. Perform basic image processing operation and convert the image into eroded image using Erode function.



7. Perform basic video processing operations on captured video. Read video using Python and display in slow motion and fast motion.

CV-Q-7.py - C:/Users/KISHORE A/AppData/Local/Programs/Python/Python313/CV-Q-7.py (3.13.12)

File Edit Format Run Options Window Help

```
import cv2

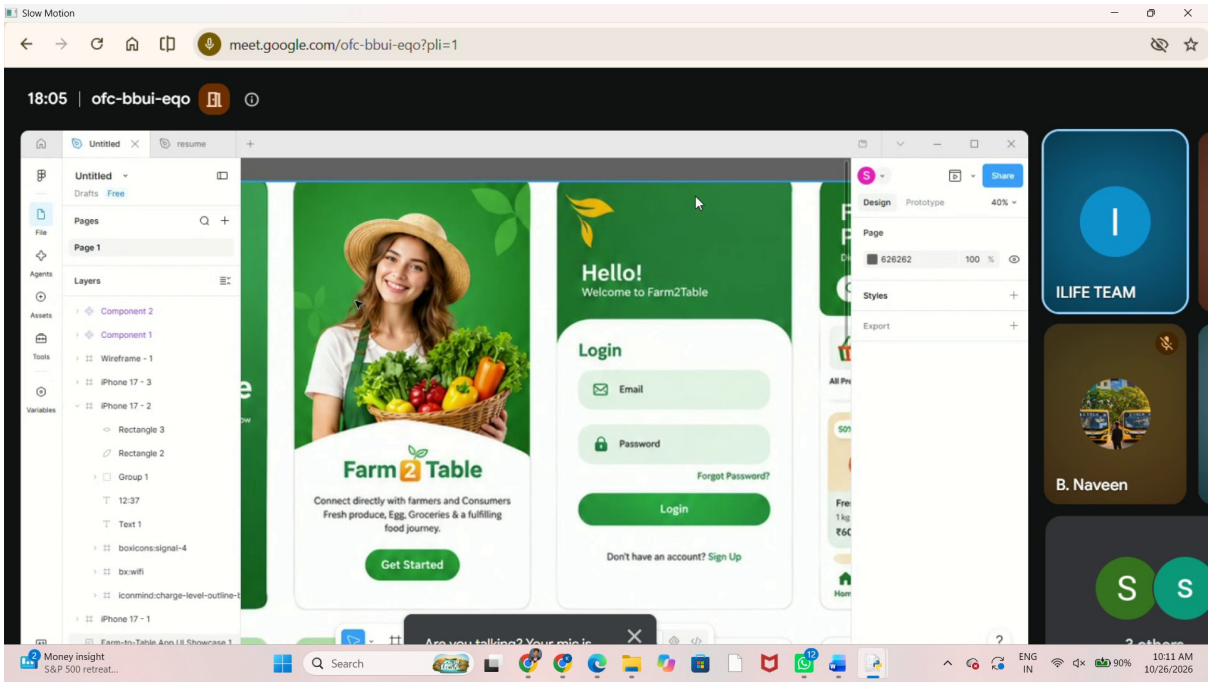
path = r"C:\Users\KISHORE A\Videos\Screen Recordings\Screen Recording 2026-10-22 182019.mp4"
cap = cv2.VideoCapture(path)

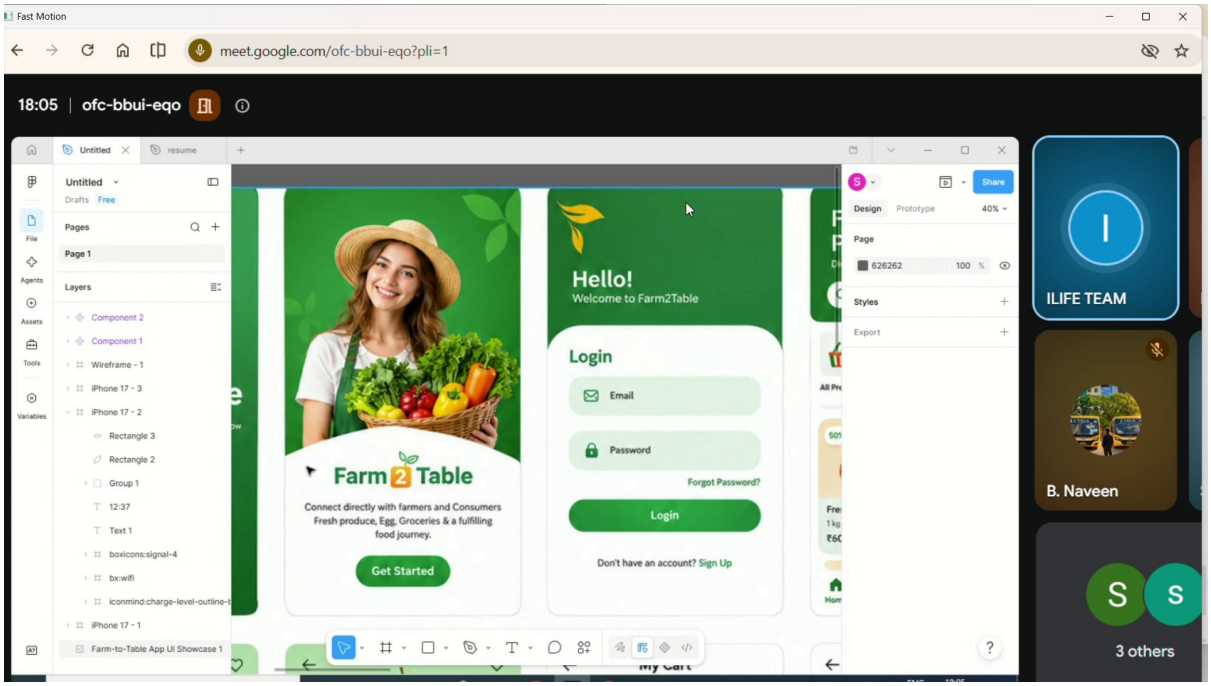
while True:
    ret, frame = cap.read()
    if not ret:
        break

    cv2.imshow("Slow Motion", frame)
    if cv2.waitKey(100) & 0xFF == ord('q'):
        break

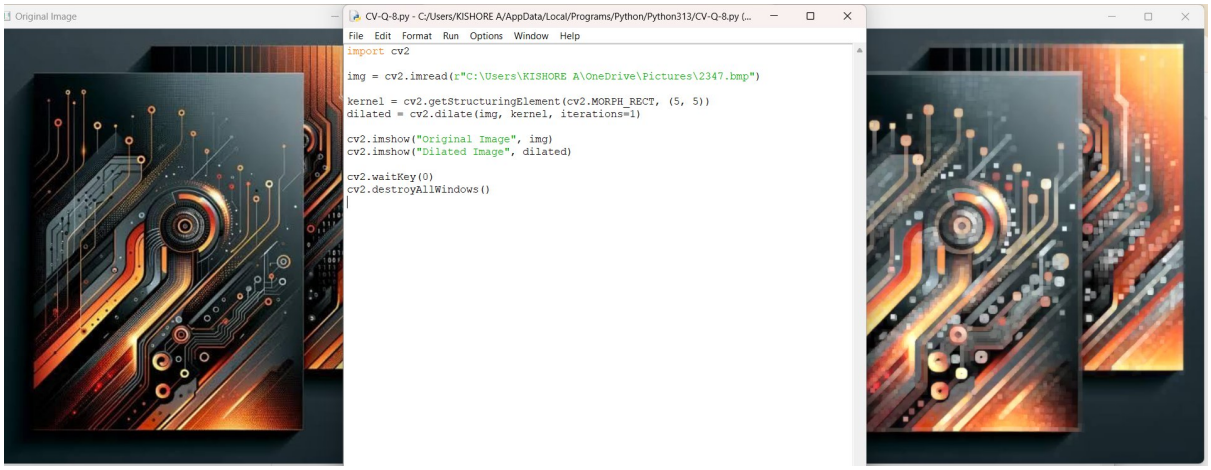
    cv2.imshow("Fast Motion", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```





8. Perform basic image processing operation and dilate an image using Dilate function.



9. Implement image scaling techniques to resize images into bigger and smaller sizes.

CV-Q-9.py - C:/Users/KISHORE A/AppData/Local/Programs/Python/Python313/CV-Q-9.py (...

File Edit Format Run Options Window Help

```
import cv2

img = cv2.imread(r"C:\Users\KISHORE A\OneDrive\Pictures\2347.bmp")

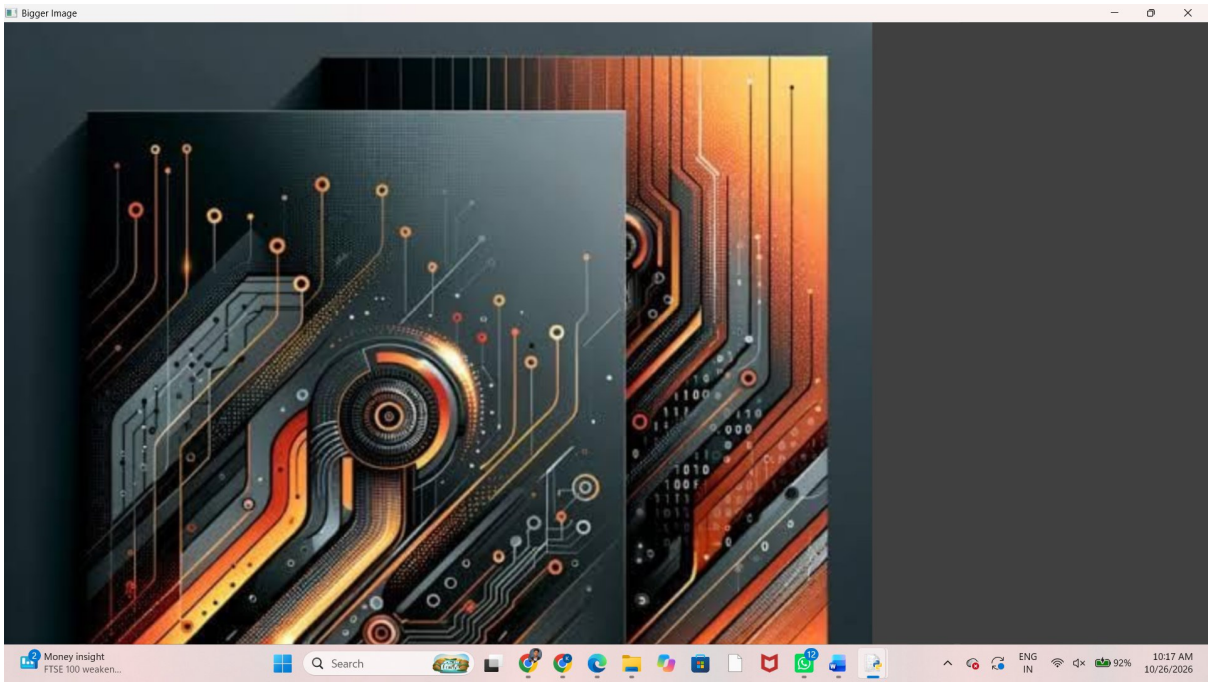
big = cv2.resize(img, None, fx=2, fy=2)
small = cv2.resize(img, None, fx=0.5, fy=0.5)

cv2.imshow("Original Image", img)
cv2.imshow("Bigger Image", big)
cv2.imshow("Smaller Image", small)

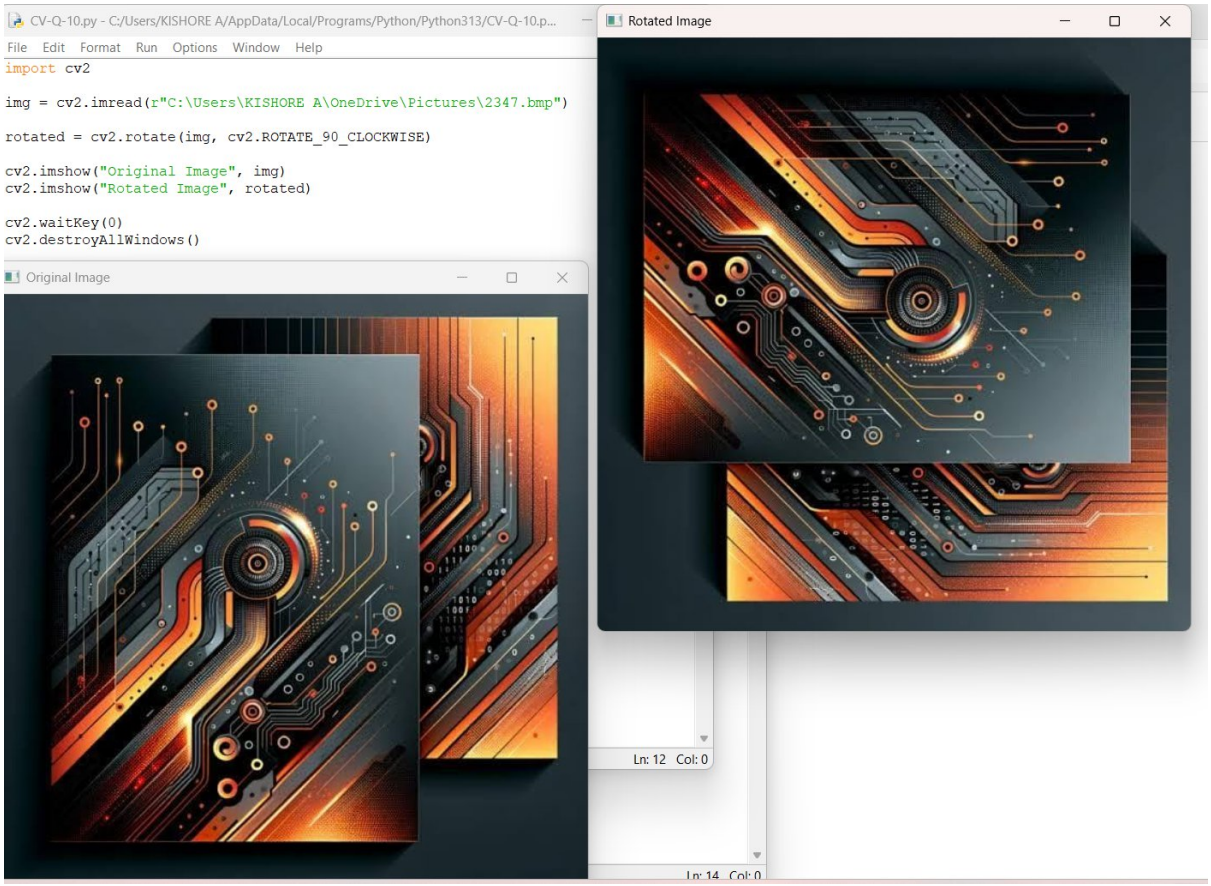
cv2.waitKey(0)
cv2.destroyAllWindows()
```



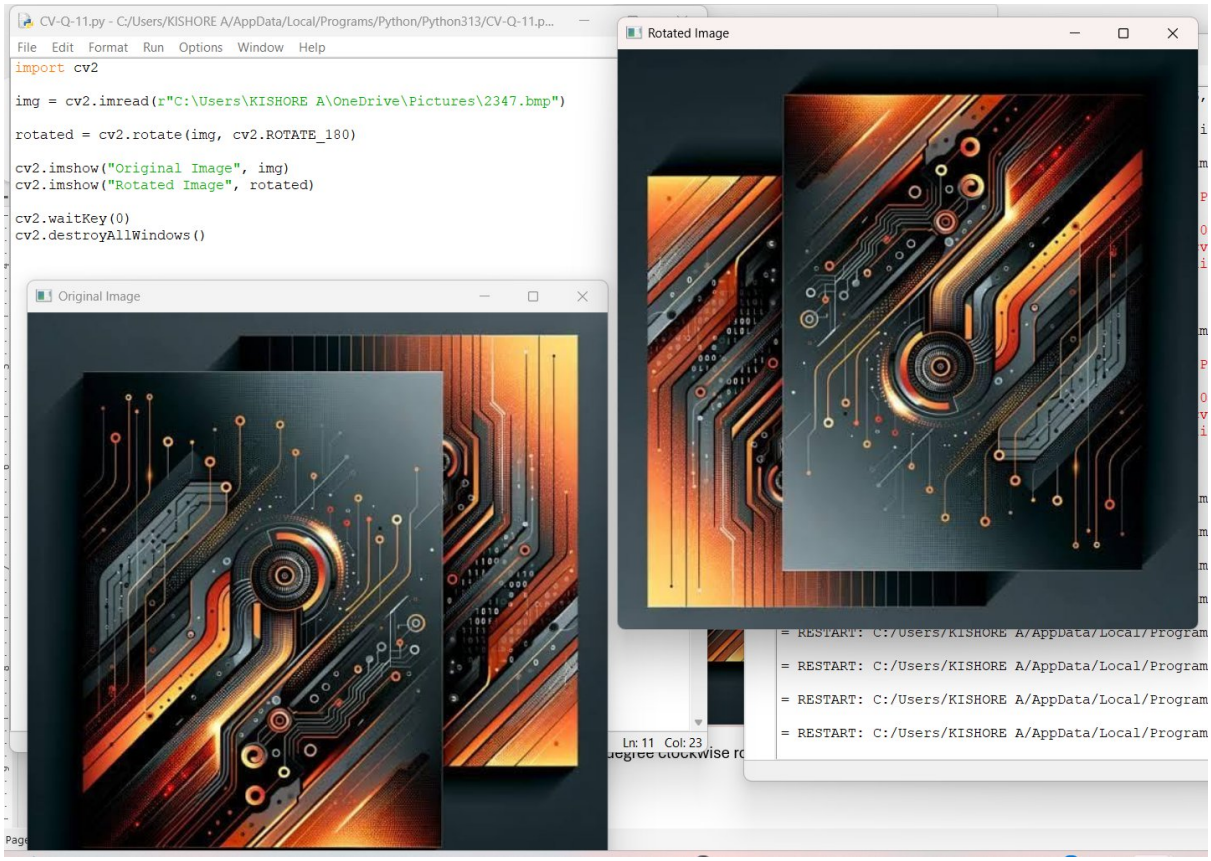

Original Image



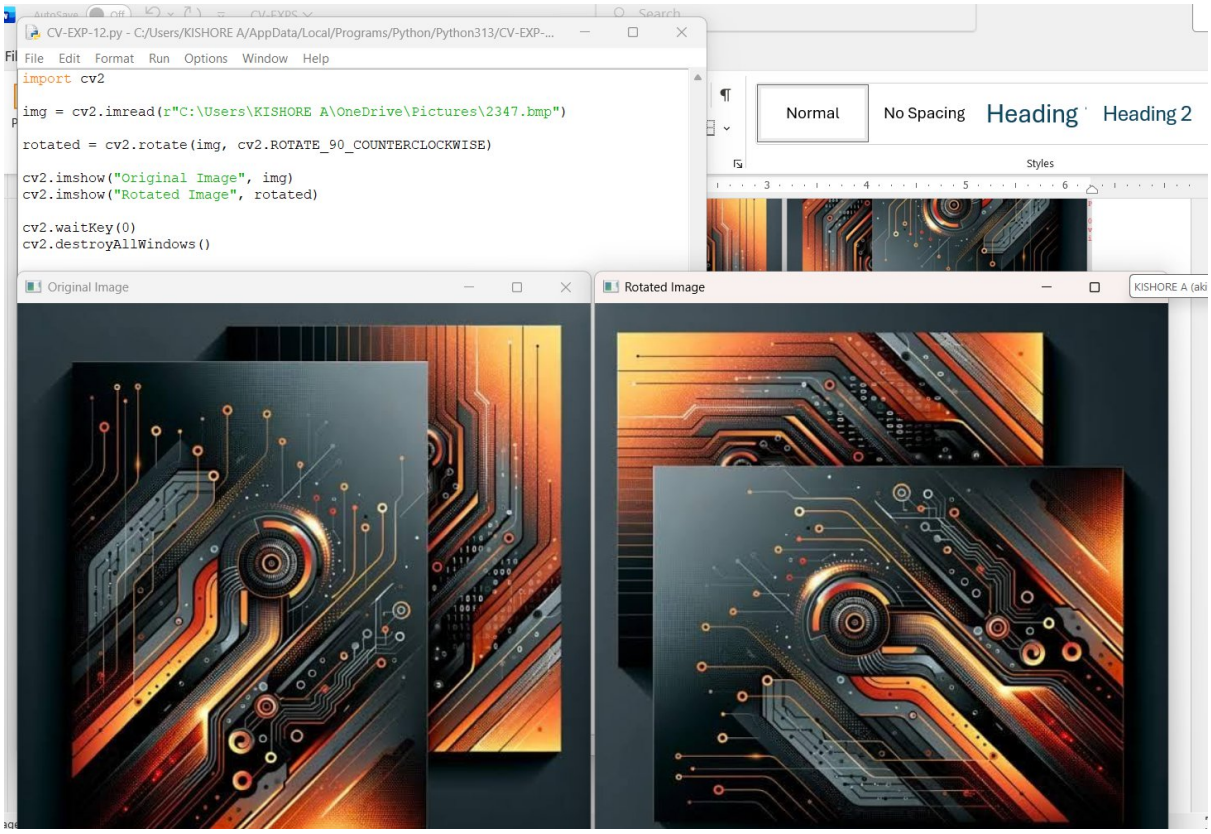
10. Perform 90-degree clockwise rotation of the given image.



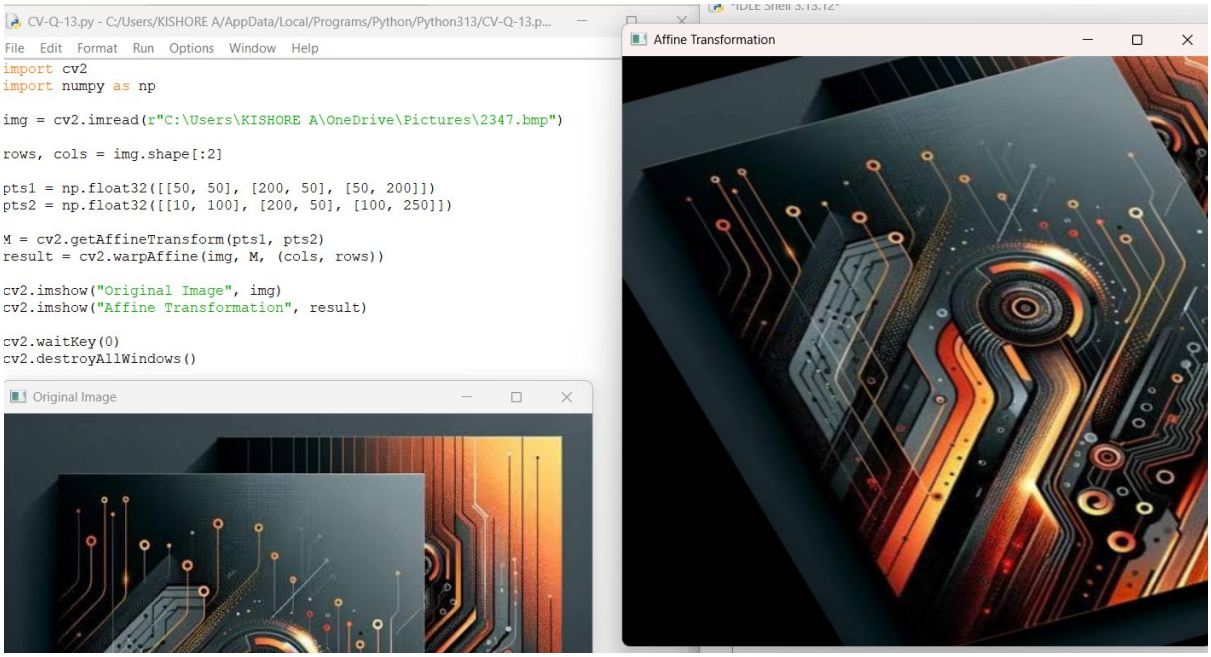
11. Perform 180-degree clockwise rotation of the given image.



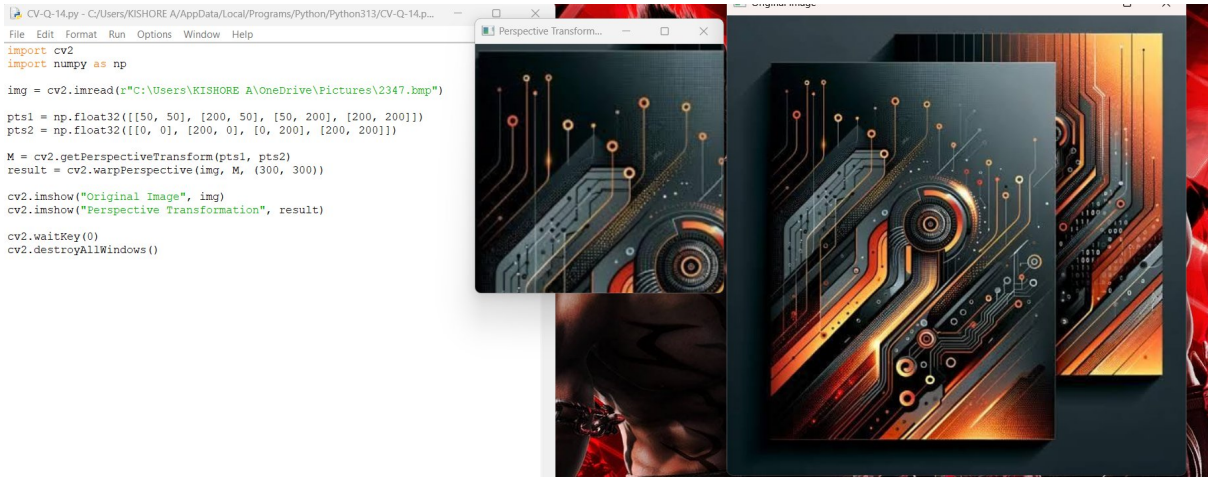
12. Perform 270-degree clockwise rotation of the given image.



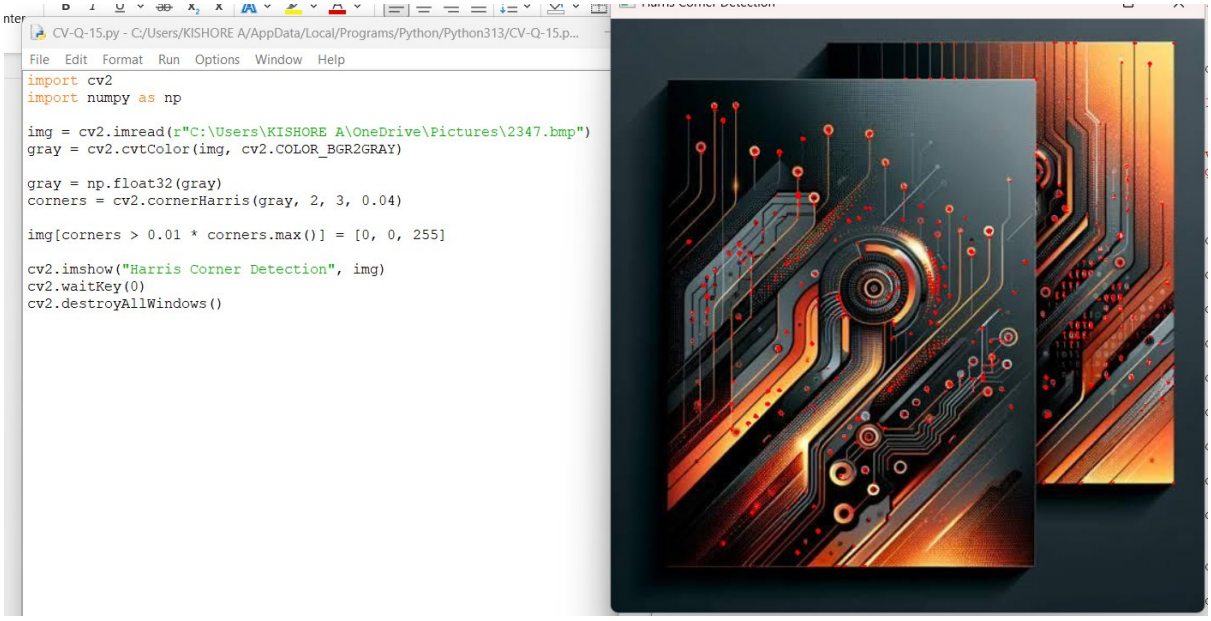
13. Perform Affine Transformation on the given image using Python and OpenCV.



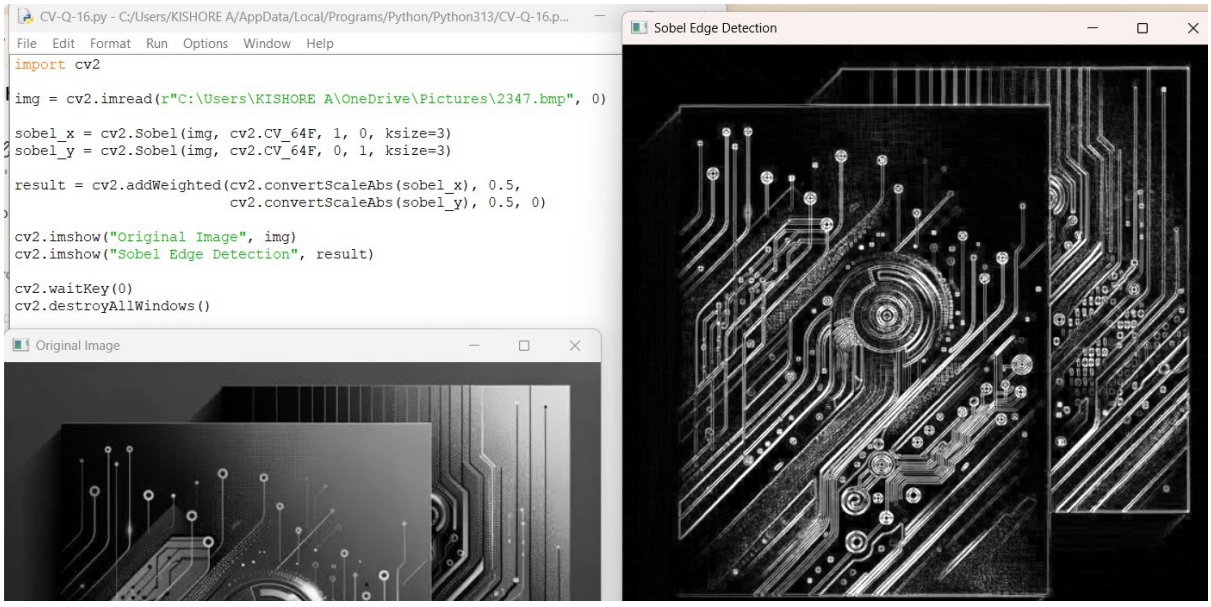
14. Perform Perspective Transformation on the given image using Python and OpenCV.



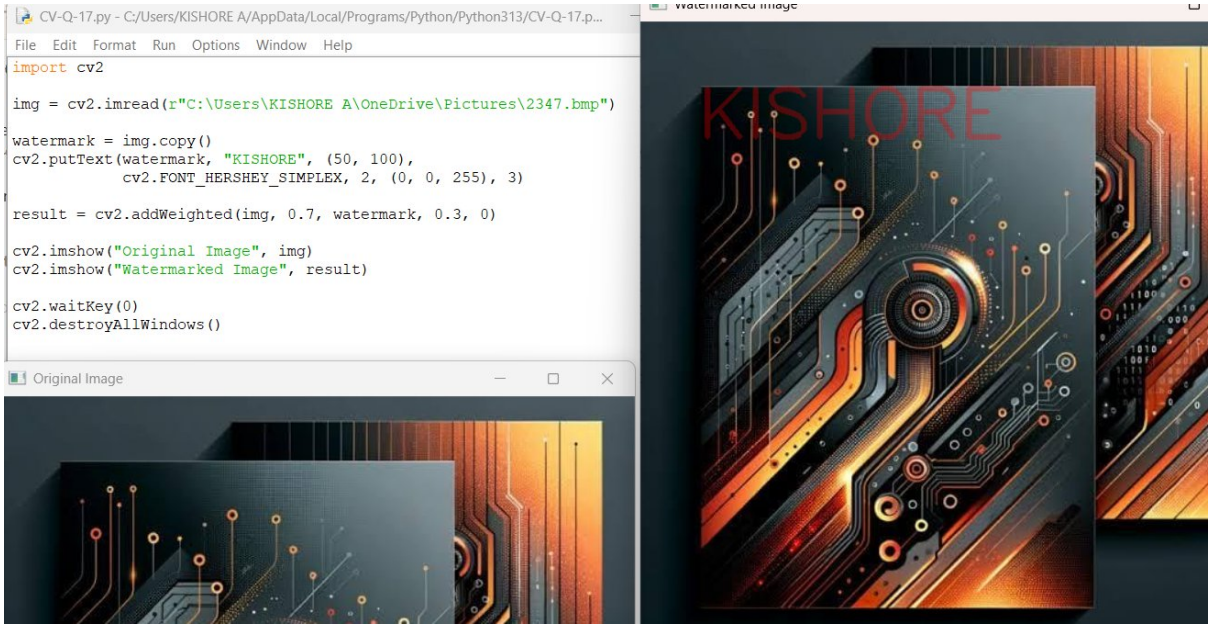
15. Detect corners in the given image using Harris Corner Detection.



16. Implement Sobel algorithm using OpenCV to filter the input image.



17. Design and implement watermarking technique to insert watermark into the original image using OpenCV.



18. Implement image cropping, copying and pasting operations to select Region of Interest (ROI) using OpenCV.

CV-Q-18.py - C:/Users/KISHORE A/AppData/Local/Programs/Python/Python313/CV-Q-18.p...

File Edit Format Run Options Window Help

```
import cv2

img = cv2.imread(r"C:\Users\KISHORE A\OneDrive\Pictures\2347.bmp")

roi = img[50:200, 50:200]  # Crop ROI
copy = roi.copy()          # Copy ROI
img[250:400, 250:400] = copy  # Paste ROI

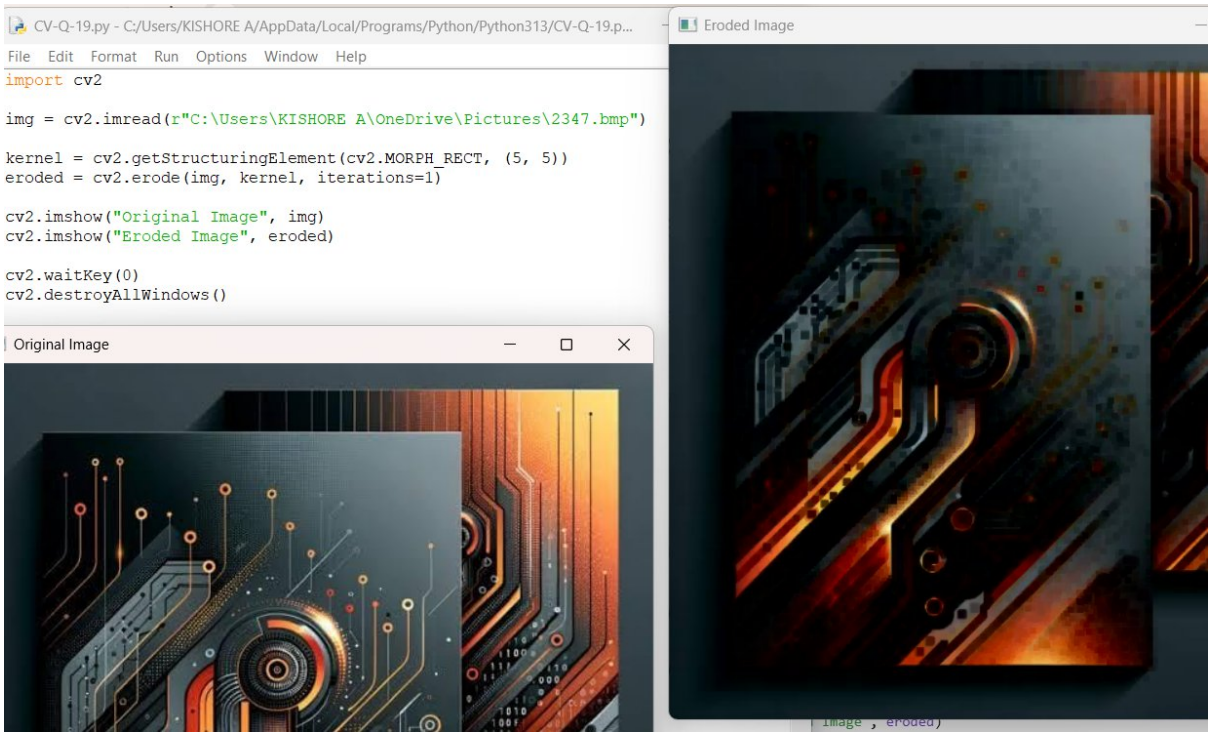
cv2.imshow("Result", img)
cv2.imshow("Cropped ROI", roi)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

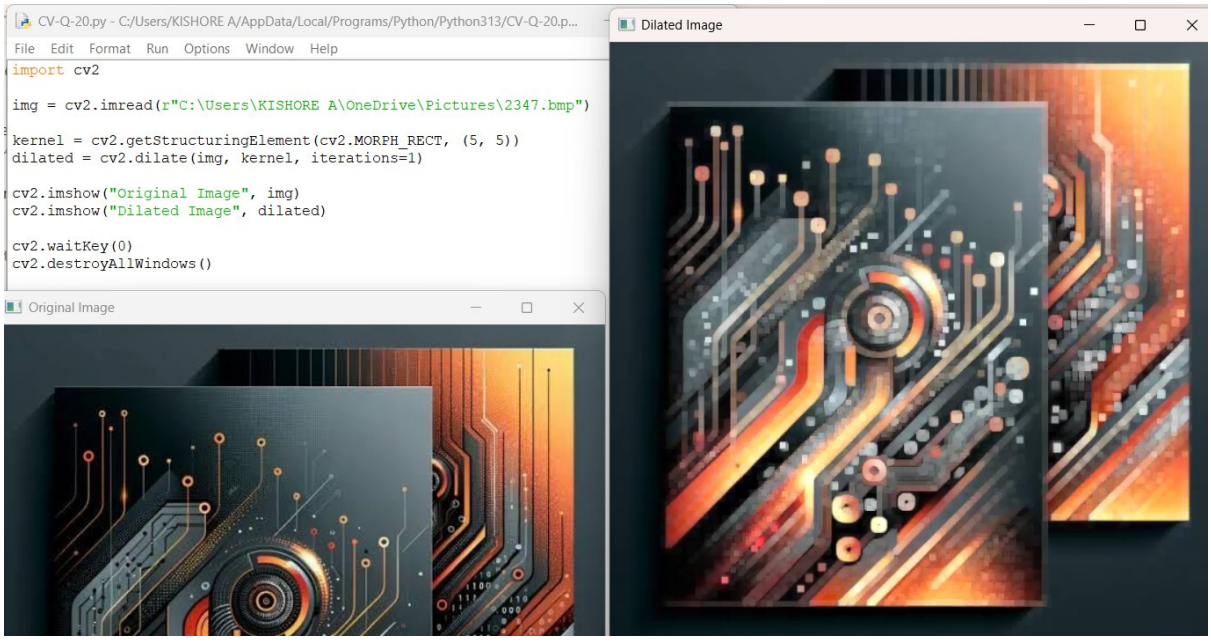




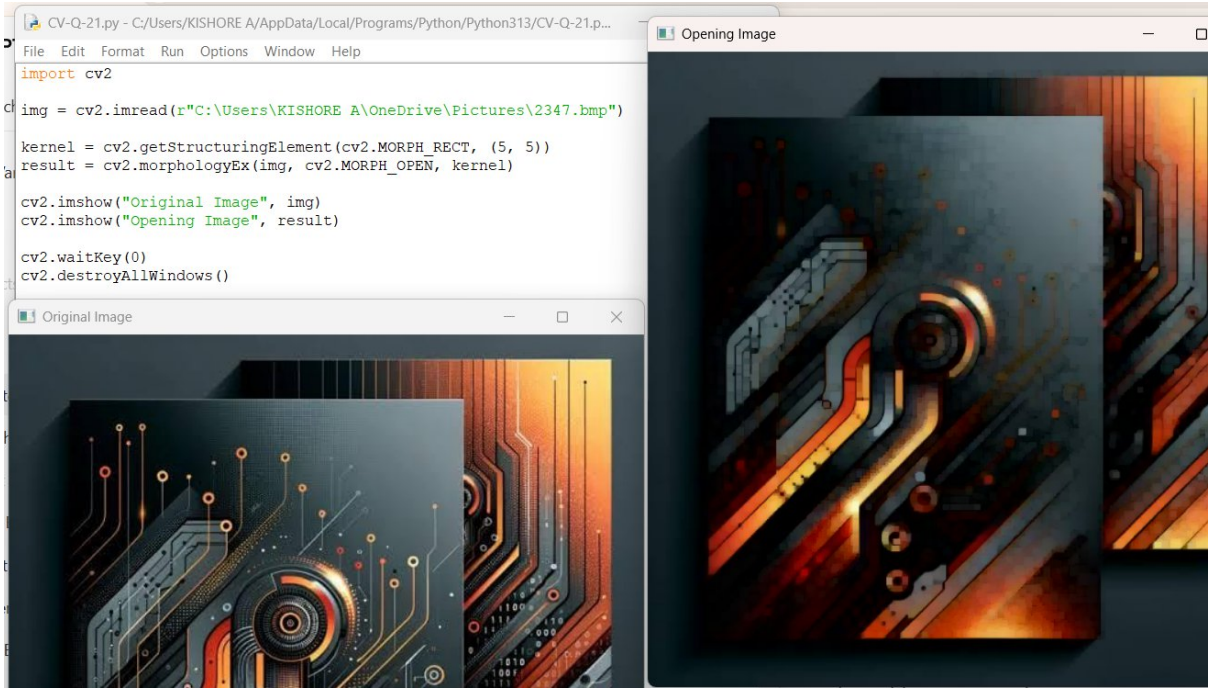
19. Implement Erosion Morphological operation using OpenCV in Python.



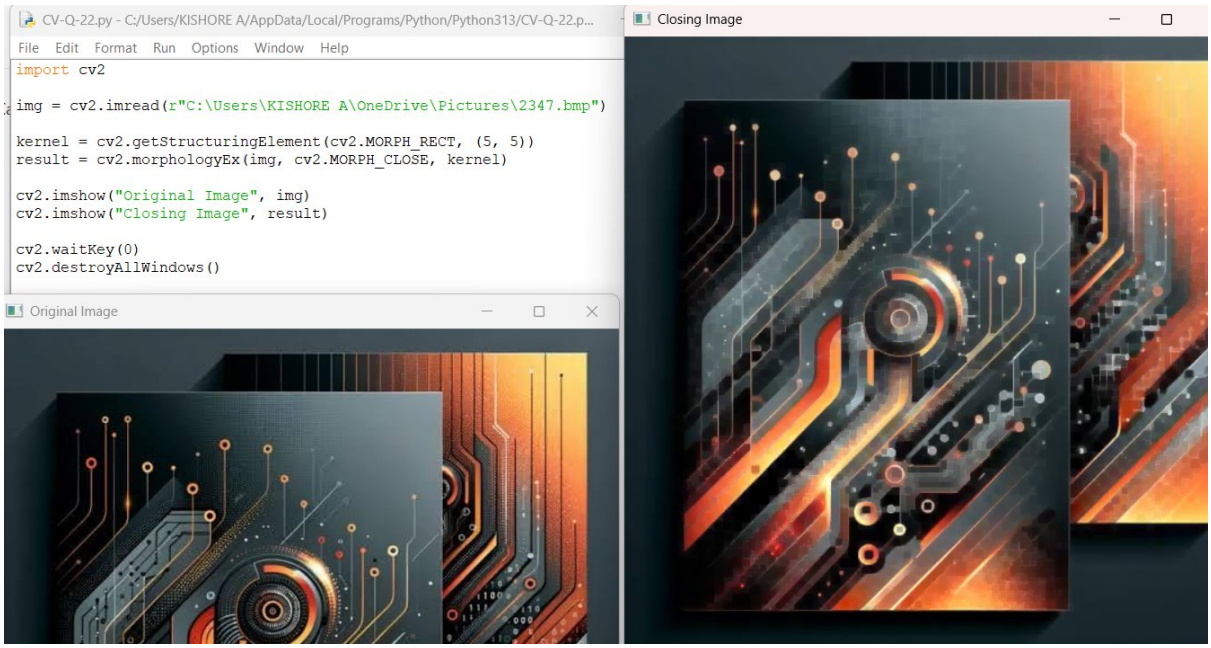
20. Implement Dilation Morphological operation using OpenCV in Python.



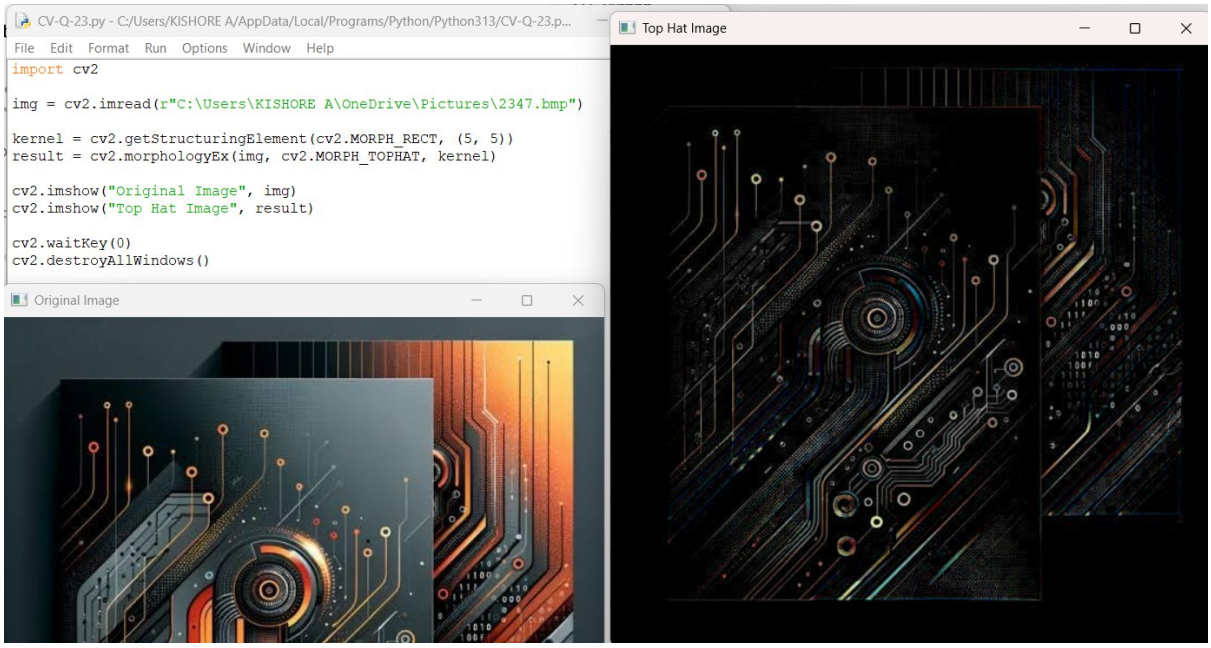
21. Implement Opening Morphological operation using OpenCV.



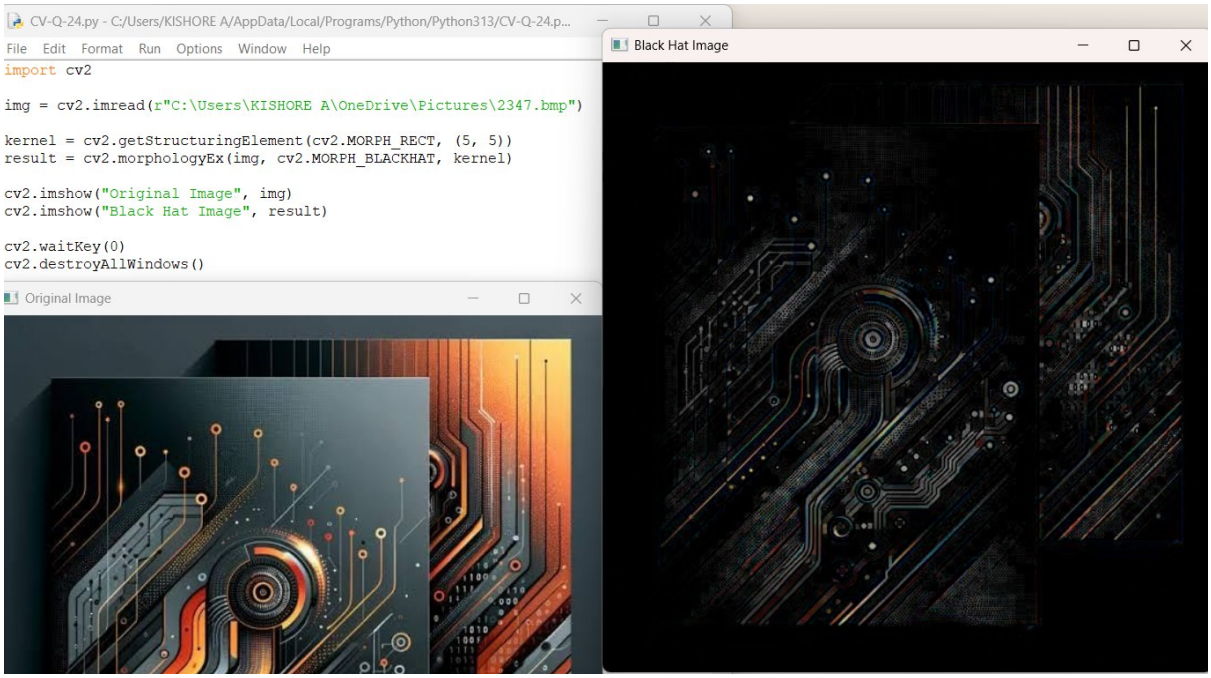
22. Implement Closing Morphological operation using OpenCV.



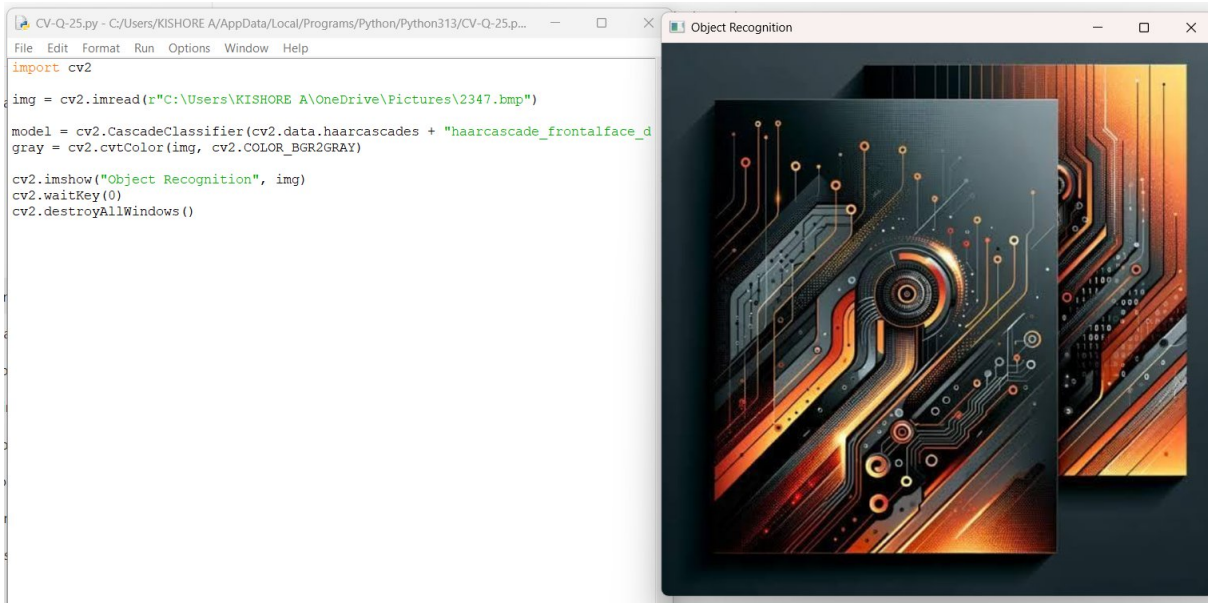
23. Implement Top Hat Morphological operation using OpenCV.



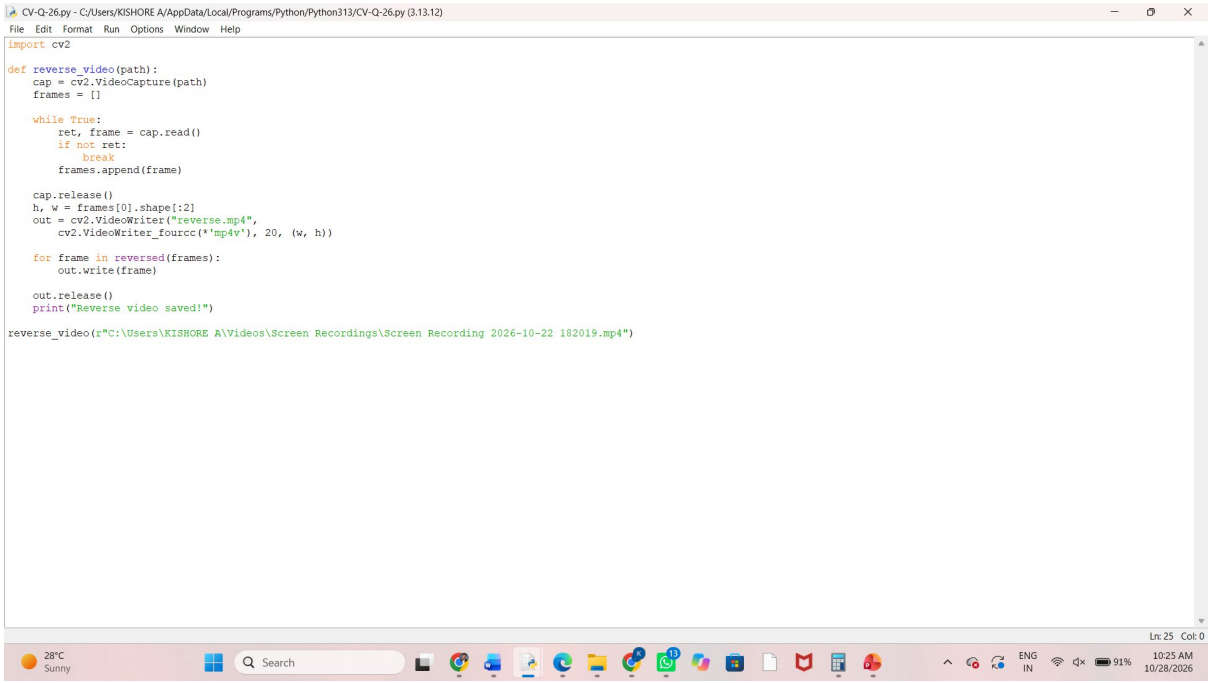
24. Implement Black Hat Morphological operation using OpenCV.



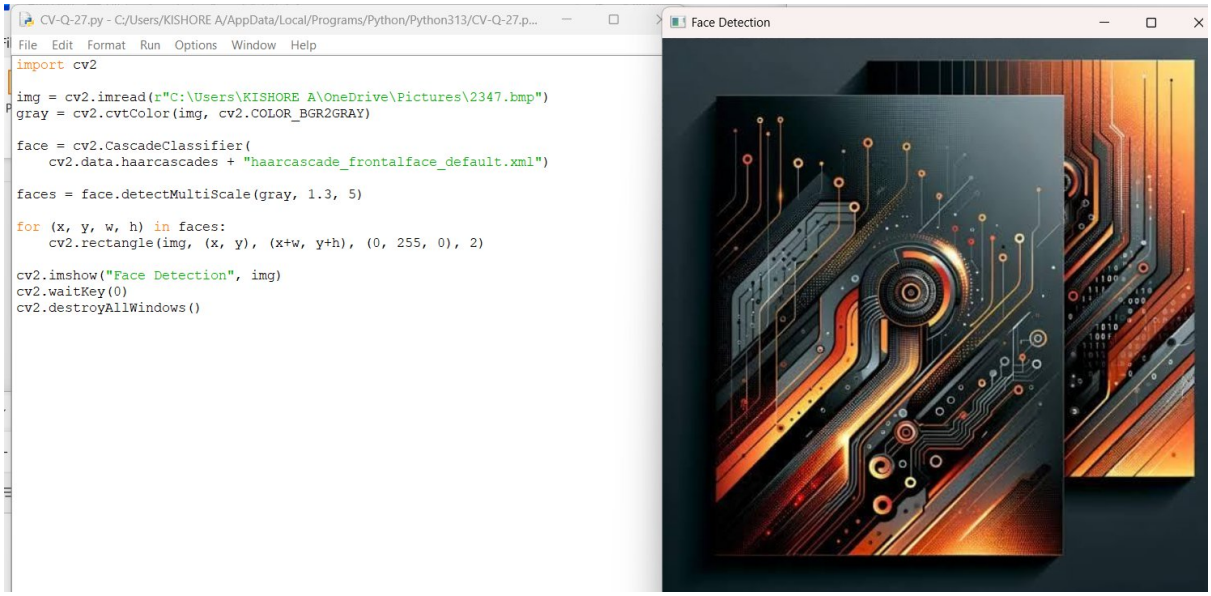
25. Recognize watch from the given image using object recognition technique.



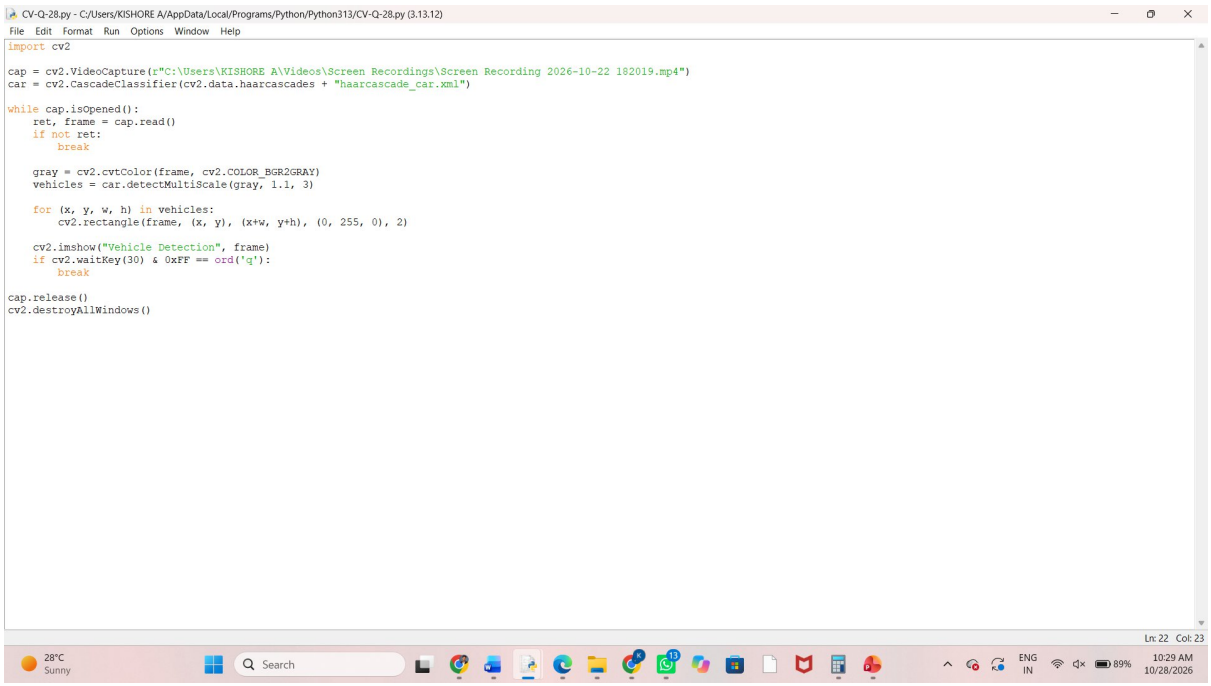
26. Implement a function to reverse frames of a video and create reverse video using OpenCV.



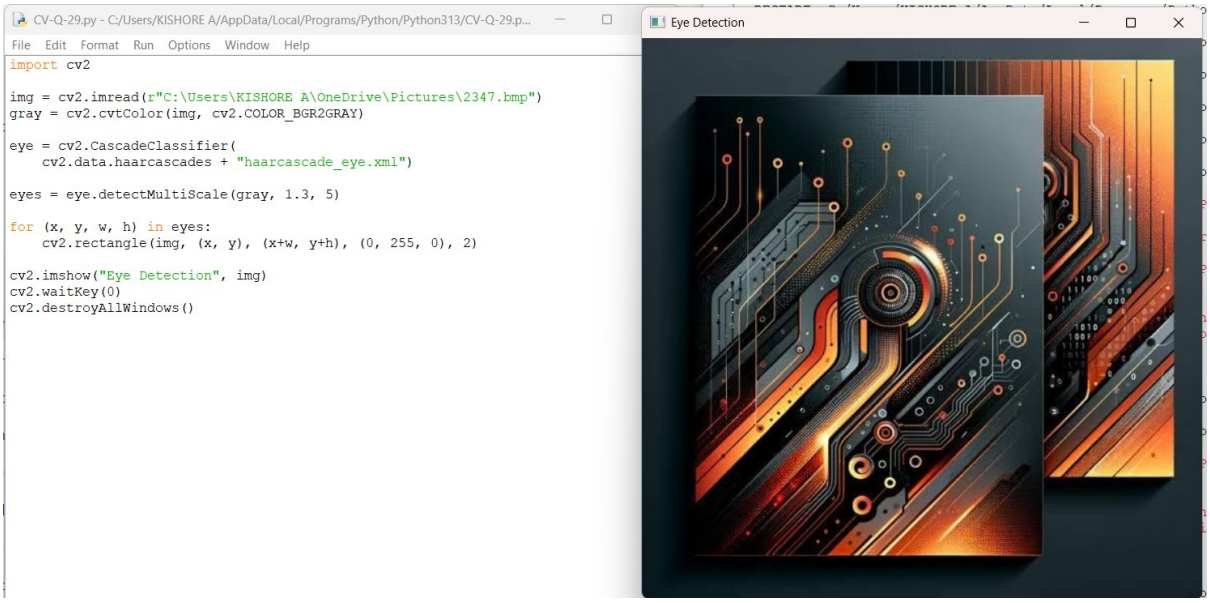
27. Implement face detection algorithm to detect and locate human faces using OpenCV.



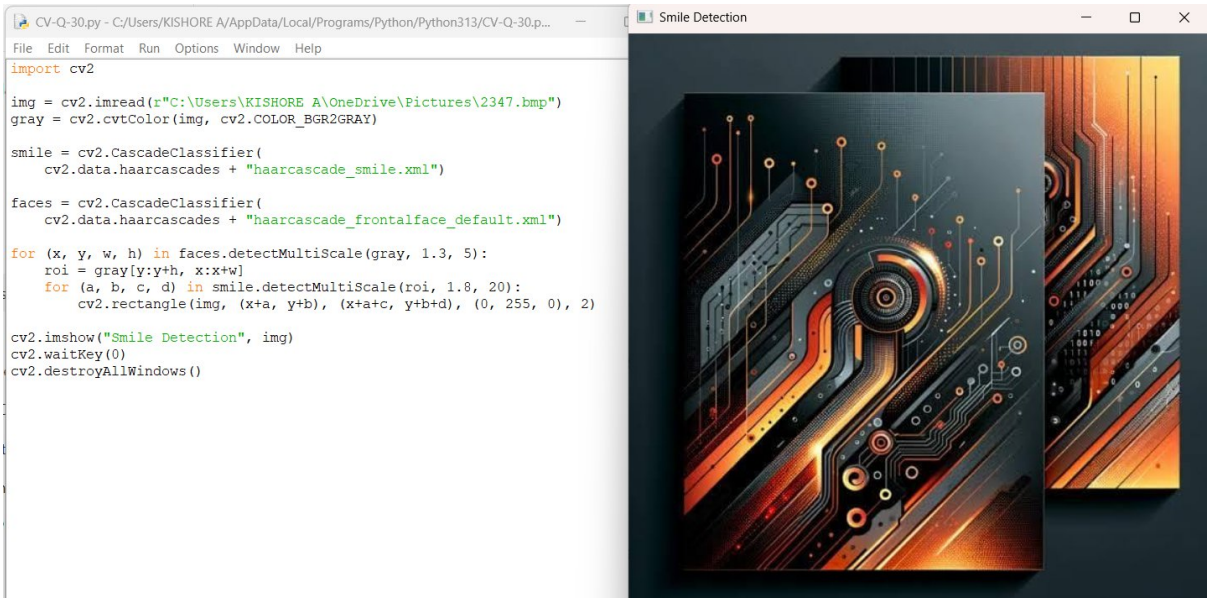
28. Implement vehicle detection algorithm to detect vehicles in each frame of video using OpenCV.



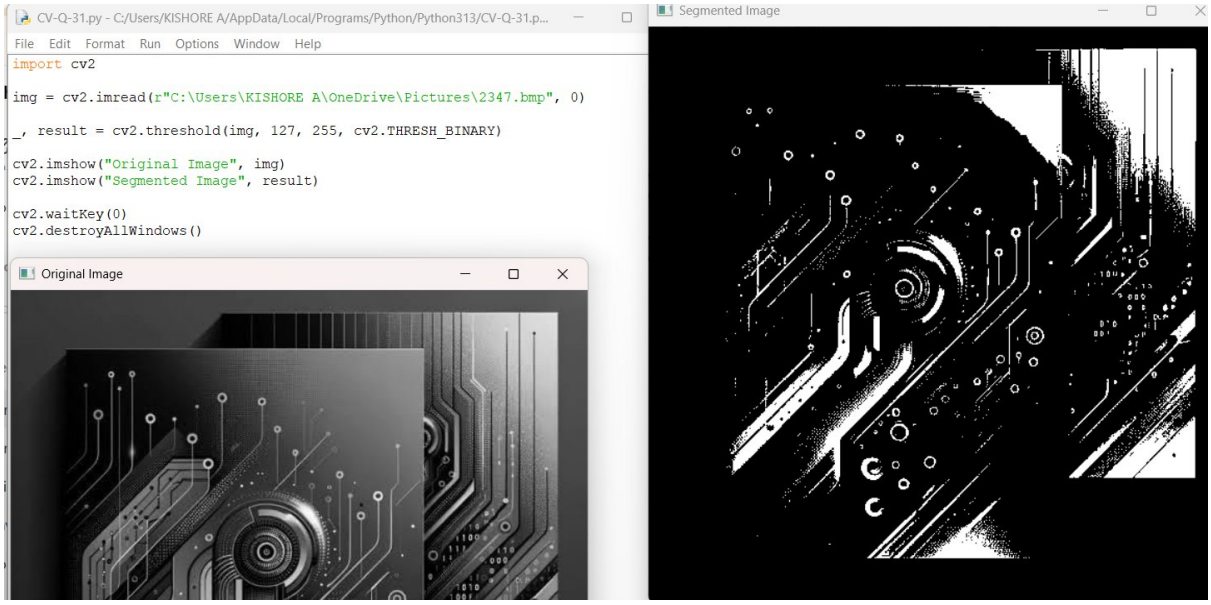
29. Implement eye detection algorithm using OpenCV.



30. Implement smile detection algorithm using OpenCV.



31. Implement image segmentation algorithm using OpenCV based on threshold values.



32. Create a white image of user-defined size and create four colored boxes (Black, Blue, Green and Red) at each corner of the image.

CV-Q-32.py - C:/Users/KISHORE A/AppData/Local/Programs/Python/Python313/CV-Q-32.p...

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

h, w = map(int, input("Enter height and width: ").split())
img = np.ones((h, w, 3), dtype=np.uint8) * 255

s = 50
img[0:s, 0:s] = (0, 0, 0)           # Black
img[0:s, w-s:w] = (255, 0, 0)       # Blue
img[h-s:h, 0:s] = (0, 255, 0)       # Green
img[h-s:h, w-s:w] = (0, 0, 255)    # Red

cv2.imshow("Colored Boxes", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

33. Create a Rectangle shape using OpenCV.

CV-Q-33.py - C:/Users/KISHORE A/AppData/Local/Programs/Python/Python313/CV-Q-33.p...

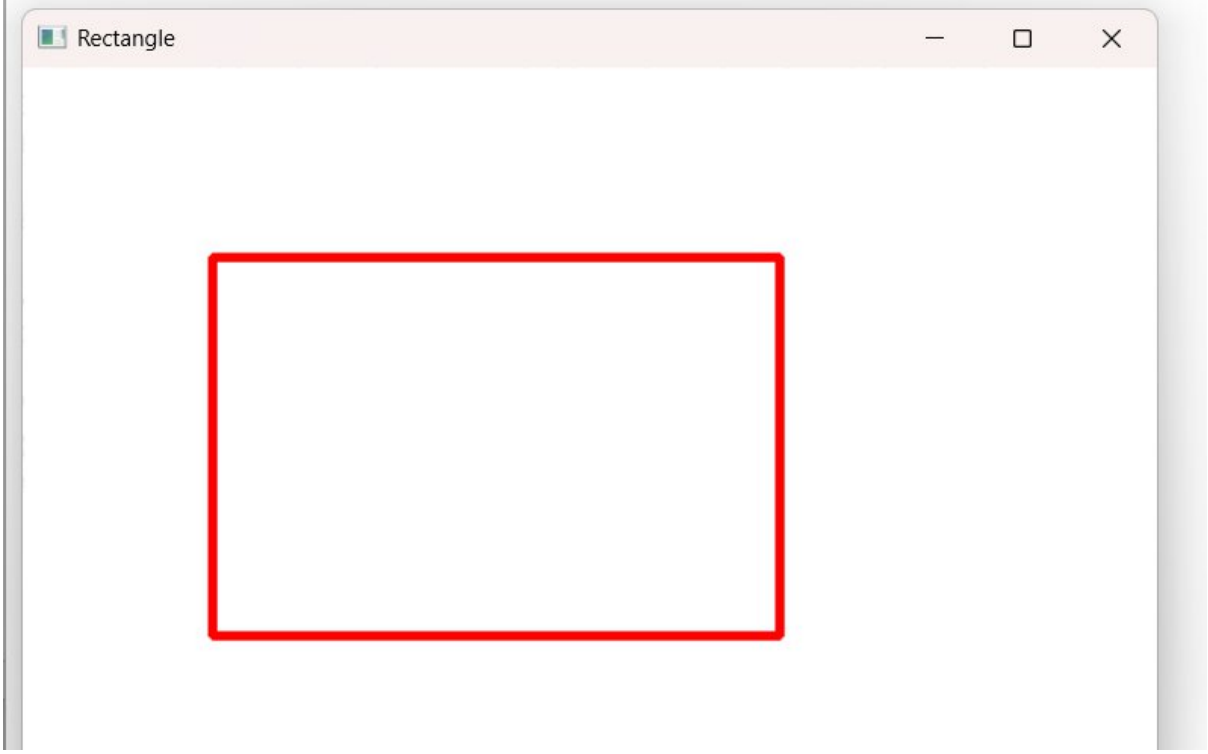
File Edit Format Run Options Window Help

```
import cv2
import numpy as np

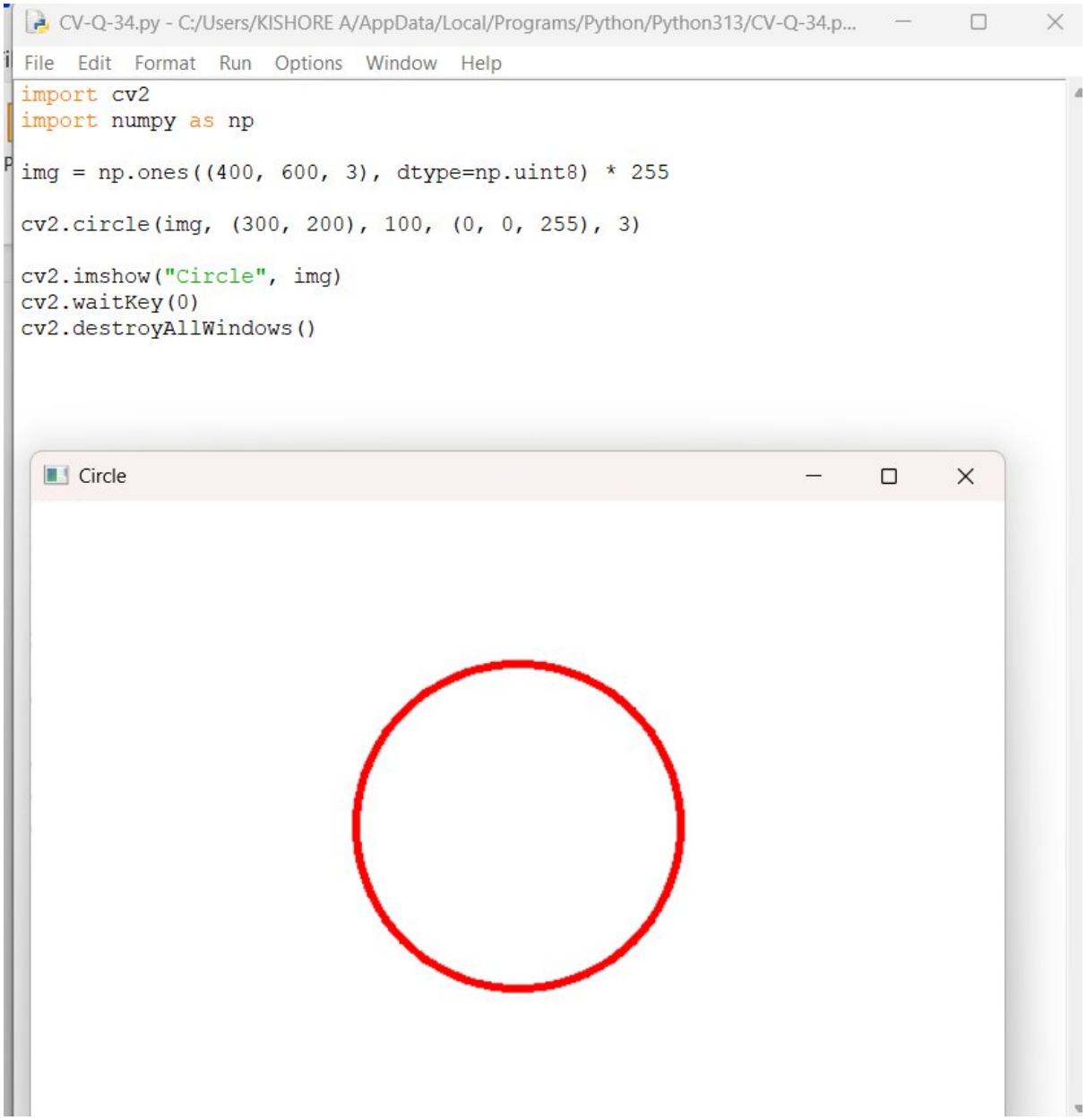
img = np.ones((400, 600, 3), dtype=np.uint8) * 255

cv2.rectangle(img, (100, 100), (400, 300), (0, 0, 255), 3)

cv2.imshow("Rectangle", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



34. Create a Circle shape using OpenCV.



35. Display user entered text string on an image using OpenCV.

CV-Q-35.py - C:/Users/KISHORE A/AppData/Local/Programs/Python/Python313/CV-Q-35.p...

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = np.ones((400, 600, 3), dtype=np.uint8) * 255
text = input("Enter text: ")

cv2.putText(img, text, (50, 200),
            cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)

cv2.imshow("Text Image", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

36. Perform background subtraction based on color levels using OpenCV.

CV-Q-36.py - C:/Users/KISHORE A/AppData/Local/Programs/Python/Python313/CV-Q-36.p...

File Edit Format Run Options Window Help

```
import cv2
import numpy as np

img = cv2.imread(r"C:\Users\KISHORE A\OneDrive\Pictures\2347.bmp")

lower = np.array([0, 0, 0])
upper = np.array([100, 100, 100])

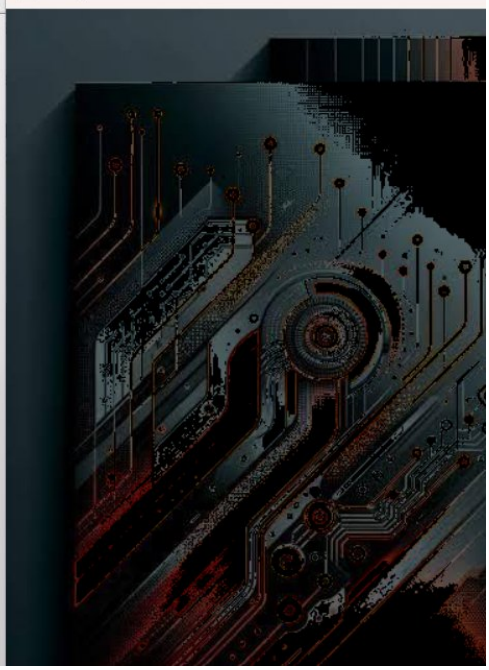
mask = cv2.inRange(img, lower, upper)
result = cv2.bitwise_and(img, img, mask=mask)

cv2.imshow("Original Image", img)
cv2.imshow("Background Subtraction", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

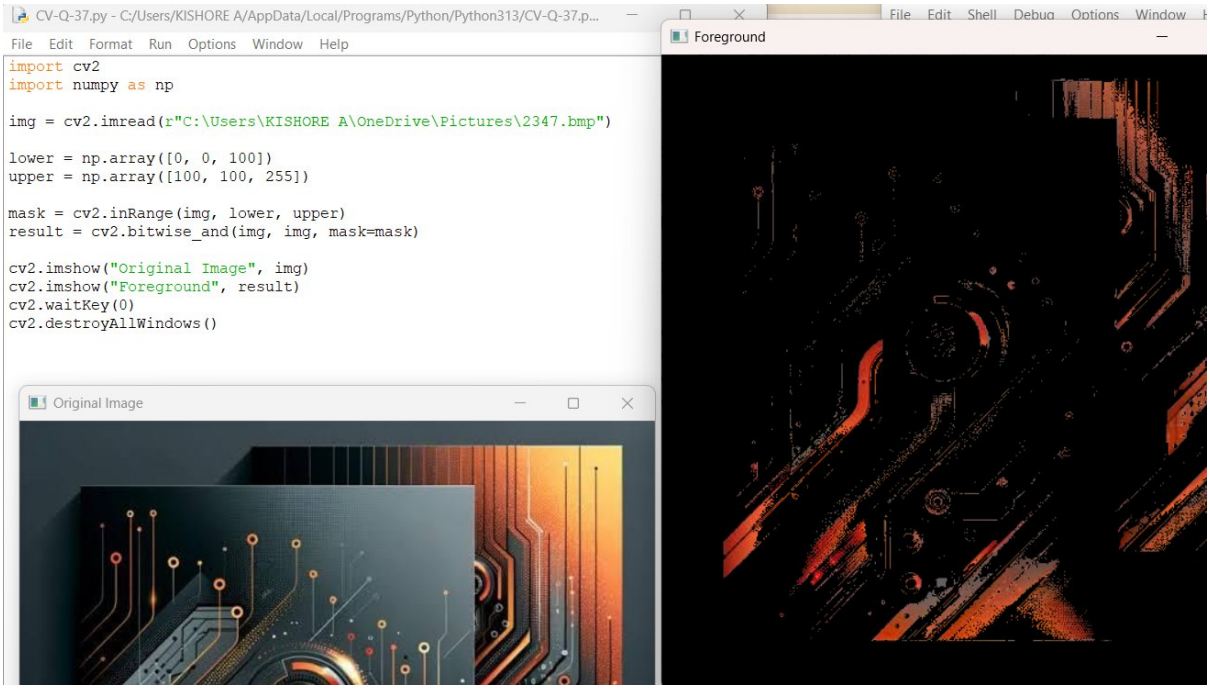
Original Image



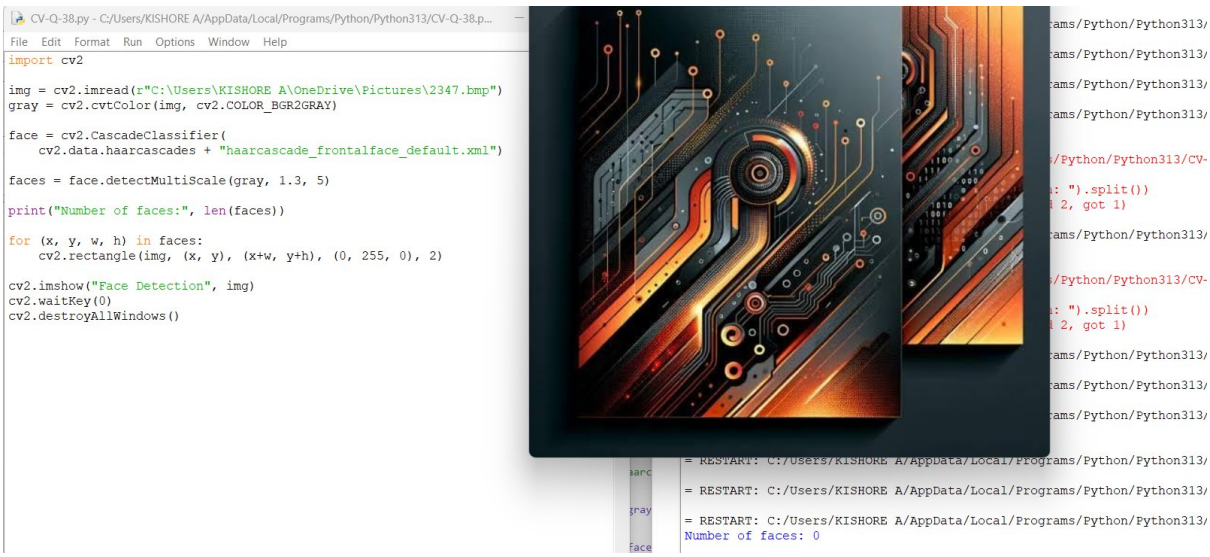
Background Subtraction



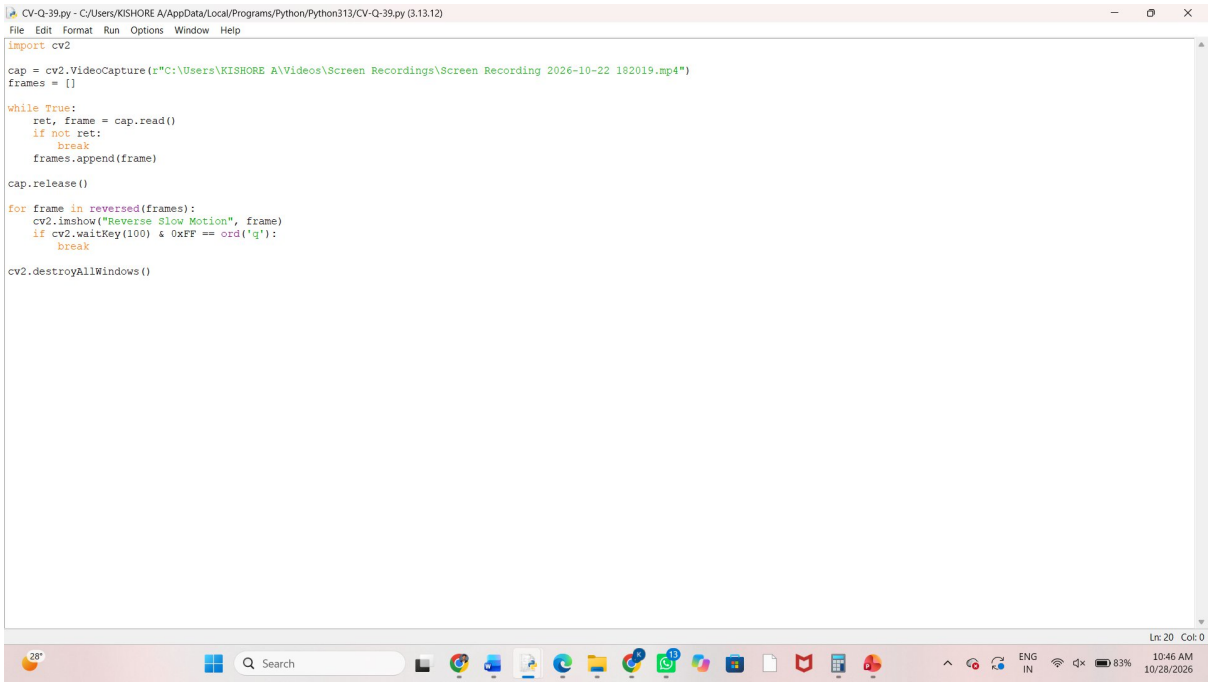
37. Perform foreground subtraction based on color levels using OpenCV.



38. Count the number of faces in the given input image using OpenCV.



39. Play the given video in reverse mode with slow motion.



40. Extract text from videos using OpenCV.

